

ATHARV SHELKE

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SUMMARY

Backend-focused Software Engineer experienced in building distributed systems, async processing pipelines, and high-performance APIs. Strong in concurrency control, data consistency, and system reliability, with proven impact on latency, cost, and fault tolerance in production systems.

EDUCATION

MGM University

Bachelor of Technology in Computer Science — CGPA: 7.8/10

Chh. Sambhajinagar, MH

Expected May 2026

TECHNICAL SKILLS

Languages: JavaScript / TypeScript, Python

Backend: Node.js, Express.js, Django, FastAPI, .NET Core

Frontend: React.js, Next.js, Tailwind CSS

Databases: PostgreSQL, MongoDB, MySQL, Redis

Tools & Auth: Git, Docker, Prisma, Vercel, OAuth2, JWT, REST

EXPERIENCE

Insourcify GmbH

Jan 2026 – Jun 2026

Backend / Full-Stack Developer Intern | Django, PostgreSQL, Microsoft Graph API, Redis, Celery

Remote

- Architected an event-driven email ingestion system using Microsoft Graph API (OAuth2 + Delta Sync); reduced full-mailbox poll overhead by **~70%** through incremental sync, enabling near-real-time ticket creation at scale.
- Owned end-to-end ticket-mapping pipeline combining regex ID extraction, email thread reconstruction, and Conversation ID correlation — eliminating **100% of manual triage** across ingested threads.
- Designed async processing infrastructure with Celery + Redis (task queues, retry backoff, scheduled jobs), improving pipeline throughput and decoupling failure domains from the synchronous request path.
- Enforced concurrency-safe ticket generation via atomic DB transactions and row-level locking (`select_for_update`), eliminating race conditions under concurrent webhook bursts.
- Drove API call and CPU overhead reduction through delta-sync caching strategies; cut redundant Graph API calls by an estimated **~60%** in steady state.

TeknoTantra Systems Pvt. Ltd.

Nov 2025 – Dec 2025

Full-Stack Developer Intern | .NET Core, React.js, PostgreSQL

Hybrid

- Designed and shipped multi-tenant REST APIs with strict RBAC isolation; enforced tenant-scoped data partitioning across all endpoints, achieving **zero cross-tenant data leakage** in QA.
- Optimized critical read paths via PostgreSQL composite indexing and query plan analysis, bringing p95 response time from **>400ms** → **<100ms** on high-traffic endpoints.
- Implemented stateless JWT-based auth with fine-grained route-level middleware, enabling independently deployable service boundaries without shared session state.
- Improved perceived frontend performance using optimistic UI patterns, reducing user-perceived latency by **~40%** on mutation-heavy workflows.

Discoverr Labs.AI

Jan 2025 – Jun 2025

Full-Stack Developer Intern | Next.js, Prisma, MongoDB

Remote

- Owned hierarchical RBAC middleware securing **5,000+** users across multi-level organizational roles; enforced sub-50ms auth checks on every protected route.
- Built a server-side real-time eligibility engine with composable filter predicates, replacing ad-hoc manual processing and cutting placement-cycle turnaround by **~65%**.
- Designed async workflow isolation to decouple third-party service failures from core operations; system maintained **99.9% uptime** during external API outages.
- Integrated AI assistant backed by contextual data pipelines, reducing repetitive support interactions by **~80%** and deflecting inbound tickets.

Freelance Full-Stack Developer

Jun 2024 – Dec 2024

Inventory Management System | Next.js, Prisma, PostgreSQL

Contract

- Engineered concurrency-safe inventory operations using atomic transactions and pessimistic locking; sustained **zero oversell incidents** under simulated 50-concurrent-user load tests.
- Reduced API latency by **80%** via parallel query execution (Promise.all batching) and selective eager loading, cutting median response time from **~400ms** to **~80ms**.
- Delivered RBAC-gated financial reporting isolating sensitive margin data to finance roles, with audit log trail for every write operation.

PROJECTS

Obsidian — AI Fitness Platform | *Next.js, TypeScript, PostgreSQL, Redis, Gemini AI*

Live | GitHub

- Built fault-tolerant AI generation pipeline with schema validation and multi-stage recovery (retry → gap-fill → model fallback); raised workout plan completion rate from **82%** → **99.8%** in production.
- Implemented ND-JSON streaming with partial buffering and disconnect recovery; reduced Time-to-First-View by **85%** and eliminated blank-screen timeouts for slow connections.
- Designed multi-model fallback (Gemini → Gemma) + Redis request deduplication; prevented thundering-herd spikes and cut LLM inference costs by **30–45%** under peak load.
- Owned concurrency-safe billing with atomic balance checks and idempotency keys, guaranteeing **zero double-spend** under parallel checkout requests.

SpecForge — Hardware ERP & E-Commerce Platform | *Next.js, TypeScript, PostgreSQL, Prisma*

Live | GitHub

- Engineered a rule-based hardware compatibility engine evaluating nested constraints (PSU wattage vs. total TDP) across **40+ entity types** in **<500ms**, backed by memoized constraint graph traversal.
- Implemented serial-level inventory tracking with TTL-based reservations and background cleanup jobs; sustained **zero oversell events** under concurrent cart sessions in stress testing.
- Designed idempotent checkout pipeline using optimistic concurrency control + unique idempotency keys, achieving **0 duplicate orders** across 1,000+ simulated retry scenarios.